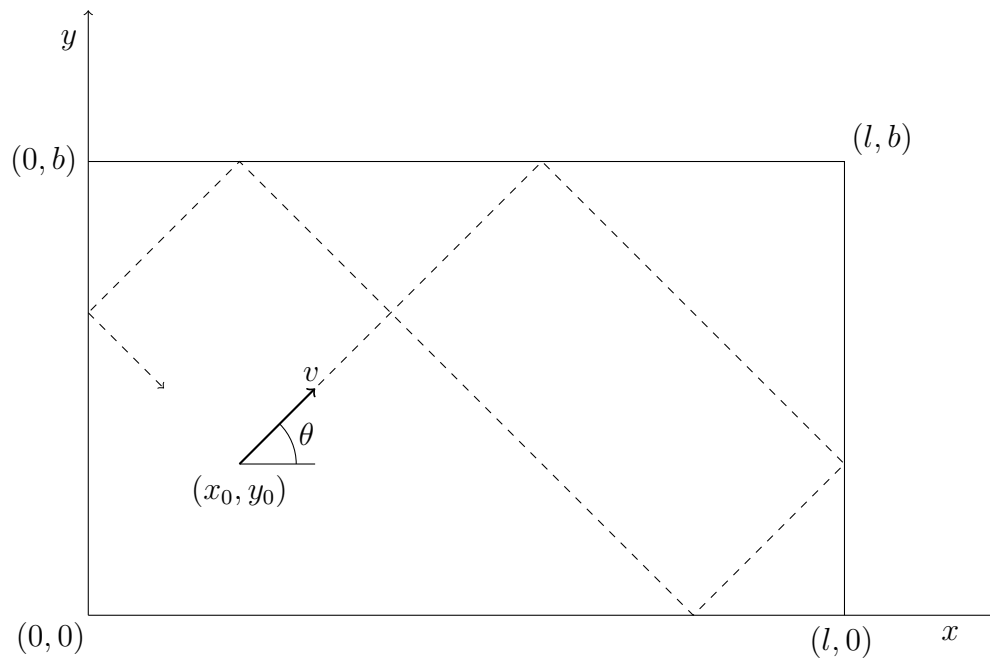


Models and Tools for Cyber-Physical Systems

Exercise sheet 11

Exercise 1: Billiards table



Consider a billiards table as schematically depicted above. We want to create a hybrid model to predict the path of the ball after being hit by the player. Assume that the table has a length l and breadth b and that the ball is laying at rest at (x_0, y_0) . A player hits the ball under angle θ giving the ball an initial speed v in that direction.

For simplicity, we assume that the ball has perfect collisions with the billiard table's sides (i.e., no energy is lost). That means that if the ball hits one of the sides parallel to the x -axis, its velocity in the y -direction flips sign while its velocity in the x -direction remains unchanged; similarly, if the ball hits one of the sides parallel to the y -direction, its velocity in the x -direction flips sign while its velocity in the y -direction remains unchanged.

- What is the initial velocity of the ball in the x -direction, $v_{x,0}$, and what is the initial velocity of the ball in the y -direction, $v_{y,0}$? Express those velocities in terms of v and θ .
- Construct a hybrid automaton capturing the motion of the ball on this perfect billiards table.

- (c) When the ball reaches one of the four corners, it drops in a hole and stops moving. Adjust your hybrid automaton to include this situation.

Exercise 2: Cruise Control

Consider the car model from the course

$$\frac{d}{dt}x(t) = v(t) \qquad \frac{d}{dt}v(t) = -\frac{k}{m}v(t) + \frac{1}{m}F(t),$$

where x is the position, v the velocity and F the force applied in case of mass m and friction coefficient k . Starting at position $x(0) = 0$, we wish to accelerate the car to the speed of 20 (m/s) using a proportional controller with appropriate gain K_p for $m = 1000$ (kg) and $k = 50$.

- Assuming that we have no feedforward control and measure only the velocity of the car, provide a continuous-time component in block diagram form.
- Create a continuous-time component of the proportional controller in block diagram form and indicate how the controller component is connected to the car model component. Finally, combine them together into a single continuous-time component.
- Using the numerical solver from the previous sheet, tune your controller, i.e., find an appropriate value for K_p , such that the car attains the speed of $v = 20$ (m/s) within $t = 100$ (s). Describe the quality of your controller in terms of steady state error, settling time, overshoot and rise time.
- Based on your observations from the pure proportional controller, design a PI controller that improves the results. Tune your chosen PI architecture.

Assume now that there is a car ahead whose position at t (s) is given by $y(t) = 150 + 15t$ (m). We wish to extend the foregoing continuous-time component to a hybrid component which

- stops accelerating as soon as the distance to the car ahead falls below 50 (m).
 - activates the previous controller once the car ahead is more than 100 (m) away.
- Create the aforementioned hybrid component in block diagram form.
 - Implement the hybrid controller by extending the numerical solver from previous sheets. By running simulations, decide whether the performance of your controller is good. If not, argue why not and try to improve it. [Hint: May it be beneficial to reset the integral error for certain mode changes?]
 - Assume now that the controller does not stop accelerating the car once the distance to the car ahead falls below 50 (m). What is the maximal force F_h with which we could accelerate the car before a crash occurs?